

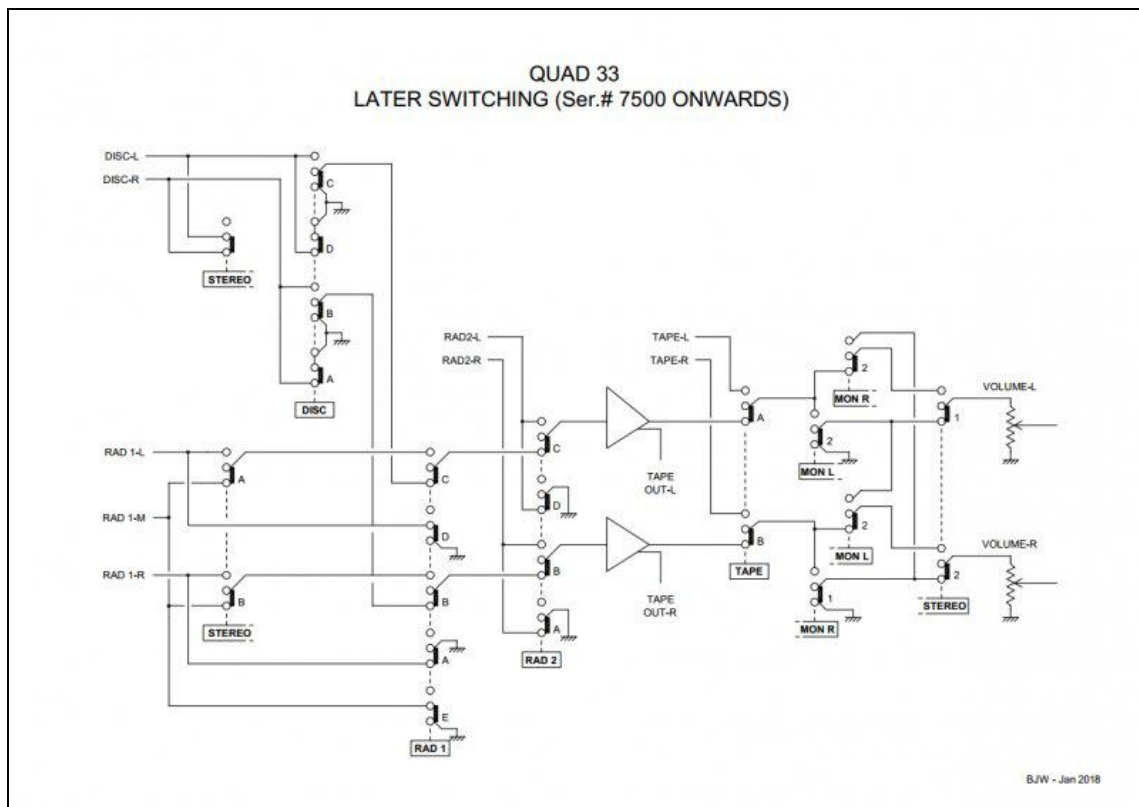
Quad 33 Control Unit Mono /Stereo switching facilities.

Mono Stereo facilities

There are a lot of questions about these facilities, in most cases these questions arise after the upgrade of a 33 Control Unit. As pointed out below, the Disc, Radio1, Radio2 and Tape replay have different mono/stereo functionalities. The only common thing is an input combined with the Stereo button. This document is based on the circuit diagram above serial number 11.500. Below serial number 7500 some functionality is missing, in the last part of this paragraph you will find the functional differences.

Also I found the switching diagrams on Vintage-Radio.net, user Trigon, hopefully I am not infringing any copyrights.

Switching Inputs above serial 7500.



First thing we noticed about this diagram, it is much clearer than the original Quad layout in the Service Documentation. Second thing we notice is the fact that the signal from the Disc or Radio1 has to pass three switches before the tape buffer, after the tape buffer there are also three switches, and then two cancel switches in the tone control circuit, and to top it off, a cancel switch in the filter stage, or three filter switches in series. No wonder the 33 is sensitive for the aging of the switch contacts. Also, when the Stereo button is not activated, the Disc input is always mono.

Summary of push-button switching

Buttons Pressed		Circuits Selected
←Mon and/or Mon→	Disc	Both channels paralleled to derive mono signal from stereo pickup and fed to left, right or both channels according to buttons pressed
Stereo	Disc Radio 2 or Tape	Each input fed separately to its own channel
←Mon and/or Mon→	Radio 1	Pin 1 of Radio 1 input fed to left, right or both channels according to buttons pressed.
Stereo	Radio 1	Pins 3 and 5 of Radio 1 input separately fed to left and right channels respectively
←Mon	Radio 2 or Tape	Pin 3 of Radio 2 or Tape inputs fed to left channel
←Mon * plus Stereo	Radio 2 or Tape	Pin 3 of Radio 2 or Tape inputs fed to both channels
Mon→	Radio 2 or Tape	Pin 5 of Radio 2 or Tape inputs fed to right channel
Mon→ * plus Stereo	Radio 2 or Tape	Pin 5 of Radio 2 or Tape inputs fed to both channels

*Note: On control units earlier than serial number 7500 combinations marked * do not apply there being no internal provision for two channel mono Radio 2 or Tape replay*

Above is the 'Switch' table from the latest Service Documentation. Not all possible combinations are mentioned, missing are the <Mon, Mon> with Radio2 and Tape, also the combination of <Mon, Mon>, Stereo and any input is not explained.

The first mentioned combination results in Stereo, you expected full mono! The second example results in channel reversal! Also, the Mon plus Stereo switching results apply to the Disc and Radio1 input.

So, to conclude, the new switching diagram did not solve the mono issue for the Radio1 (pin 3 & 5), Radio2 and Tape input.

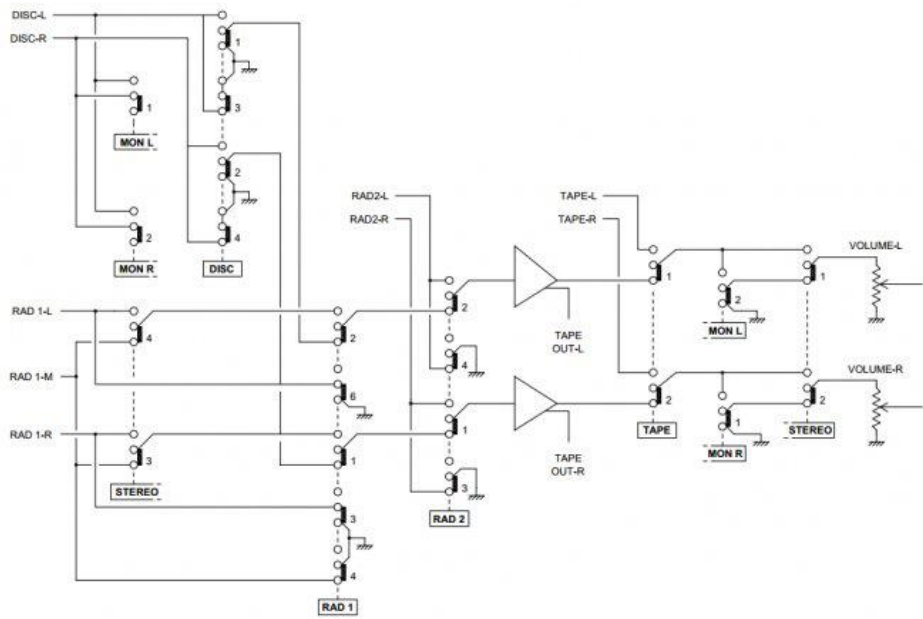
Mono on Radio1 means it is only working on tuners with a pin 1 connection.

The Left or Right switching to both outputs was (is) only of interest for two track tape machines with mono recordings on both tracks. It seems to me that also in the sixties, this was a minority group.

Why is it important to have a mono facility? With a Mono signal it is easier to centre the soundstage with the balance control.

Also I don't like the Stereo/Mono switching depending on the type of input.

QUAD 33
EARLIER SWITCHING (UP TO Ser.# 7500)



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